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Women in Charge: Evidence from Hospitals

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1 Big picture

- **Question.** Do female CEOs manage organizations differently, and are they evaluated differently by boards?
- **Setting.** The paper studies U.S. nonprofit hospitals, an unusually useful setting because female CEOs are far more common than in public firms and the data cover hospital operations, finances, governance, CEO pay, and CEO turnover.
- **Main decision-making result.** There is no evidence that female hospital CEOs choose more conservative financial policies, lower investment, more stakeholder-friendly labor policies, or more treatment of vulnerable patients.
- **Crisis test.** During the 2008 financial crisis, matched hospitals led by female and male CEOs respond almost identically. Profitability, investment, employment, salaries, and Medicaid patient shares move similarly.
- **Compensation result.** Female CEOs earn less. The raw pay gap is about 32%; after controls and hospital fixed effects it remains around 7.8%.
- **Incentive result.** Pay-for-performance sensitivity is flatter for female CEOs. Moving from the worst to best profit quintile raises male CEO pay much more than female CEO pay.
- **Turnover result.** Turnover-performance sensitivity is steeper for female CEOs. Poor performance is followed by larger increases in turnover probability for women than for men.
- **Governance interpretation.** Boards do not appear to reward female CEOs as much for good performance, but they appear to punish poor performance more strongly.
- **Tension.** The paper does not find a distinct female management style, yet boards behave as if CEO gender changes how performance signals are interpreted.
- **Economic takeaway.** The main gender asymmetry is not in managerial choices; it is in labor-market outcomes for CEOs: pay, incentives, and turnover.

2 Data and measurement

2.1 Hospital sample

- The data come from IRS filings and hospital-level information for nonprofit hospitals.
- The baseline sample contains 1,981 hospitals and 25,762 hospital-years from 2000 to 2018.
- The sample includes 4,353 CEOs, of which 819 are female.
- The paper requires the CEO to be in office for at least one year and to have nonmissing gender.
- CEO gender is identified manually using CEO names and hospital filings.

Interpretation: The sample is much richer than public-company CEO samples because female CEOs are frequent enough to permit meaningful tests within one industry.

2.2 Hospital policy variables

- **Financial conservatism:** leverage, cash/assets, securities/assets.
- **Investment and growth:** investment, service revenue growth.
- **Stakeholder orientation:** salaries, personnel growth, Medicaid ratio, Medicare ratio, charity care, contributions, and patient mix.

- **Hospital environment:** system membership, local market density, competition rank, and HSA characteristics.

Interpretation: These variables map directly into hypotheses from the gender literature: higher risk aversion should show up in safer financing or lower investment; stronger benevolence should show up in labor or patient-mix choices.

2.3 CEO pay and turnover variables

- CEO pay is measured in thousands of dollars from Form 990 compensation disclosures.
- Turnover is measured as CEO departure; "forced" turnover focuses on departures of CEOs age 60 or younger.
- Financial performance is measured by profit margin, ranked by year into quintiles.
- Pay and turnover tests include hospital controls, year effects, hospital fixed effects, CEO tenure, CEO age, and multiple-position indicators where appropriate.

Interpretation: The pay tests ask whether female CEOs are paid differently, while the performance-sensitivity tests ask whether boards translate performance into rewards and punishments differently by gender.

2.4 Financial crisis matched sample

- The crisis sample matches hospitals led by female CEOs in 2007 to hospitals led by male CEOs.
- Matching variables include revenues, density rank, investment, revenue growth, and system membership.
- The final matched sample has 134 treated hospitals and 271 control hospitals.

Interpretation: The crisis setting is valuable because the shock is plausibly unexpected and large. It tests whether gender differences appear precisely when the stakes are high.

2.5 Boards and CEO background data

- Governance data include the gender composition of hospital boards and female representation among non-CEO executives.
- CEO background data include education, internal promotion, prior title, and prior occupational background.
- These data are collected manually on smaller subsamples.

Interpretation: These sections probe mechanisms: whether female leaders are associated with female boards, whether female boards close the pay gap, and whether background differences explain labor-market outcomes.

3 Empirical specifications

3.1 Hospital characteristics and CEO gender

A baseline selection-style regression relates CEO gender to hospital attributes:

$$\text{FemaleCEO}_{it} = \alpha_t + \beta X_{it} + \varepsilon_{it}, \tag{1}$$

where X_{it} includes hospital financing, investment, performance, patient mix, or local-market attributes.

Interpretation: This is not a causal regression. It asks whether women tend to lead hospitals with particular risk profiles or stakeholder profiles. If female CEOs systematically prefer safer or more benevolent institutions, the matching pattern should reveal it.

3.2 Within-hospital changes in policies

A complementary specification includes hospital fixed effects:

$$Y_{it} = \alpha_i + \tau_t + \beta \text{FemaleCEO}_{it} + \gamma' X_{it} + u_{it}. \tag{2}$$

Interpretation: This specification asks whether a hospital changes policy when a woman becomes CEO or when a woman leaves the CEO role. It is closer to a management-style test than the cross-sectional matching regressions.

3.3 Financial crisis difference-in-differences

For matched hospitals around the financial crisis, the core regression is:

$$Y_{it} = \alpha_i + \tau_t + \beta(\text{FemaleCEO}_i \times \text{PostCrisis}_t) + u_{it}, \tag{3}$$

where PostCrisis_t equals one for 2009 through 2011.

Interpretation: If female CEOs respond differently to crises, β should be economically meaningful. The paper finds coefficients close to zero across outcomes.

3.4 Crisis event-study version

The figure-based event study replaces the single post-crisis interaction with year interactions:

$$Y_{it} = \alpha_i + \sum_k \beta_k (\text{FemaleCEO}_i \times 1\{t = k\}) + u_{it}. \tag{4}$$

Interpretation: The event study shows the timing: whether male- and female-led hospitals diverge before, during, or after the shock. The estimates are centered near zero.

3.5 CEO pay regressions

The gender pay gap is estimated using:

$$\log(Pay_{it}) = \alpha_i + \tau_t + \beta \text{FemaleCEO}_{it} + \gamma' X_{it} + u_{it}. \tag{5}$$

Interpretation: The coefficient on FemaleCEO_{it} measures the residual pay gap after comparing CEOs within similar hospitals and, in the strictest specification, within the same hospital over time.

3.6 Pay-for-performance sensitivity

Pay-for-performance sensitivity is estimated using performance quintiles:

$$\log(Pay_{it}) = \alpha_i + \tau_t + \sum_{q=2}^{5} \delta_q \text{PerfQuintile}_{q,it} + \sum_{q=2}^{5} \theta_q (\text{FemaleCEO}_{it} \times \text{PerfQuintile}_{q,it}) + \gamma' X_{it} + u_{it}. \quad (6)$$

Interpretation: The interaction coefficients θ_q test whether female CEOs receive weaker rewards for the same financial performance rank.

3.7 Turnover-performance sensitivity

CEO turnover is modeled with a probit specification:

$$\Pr(Turnover_{it} = 1) = \Phi\left(\alpha + \beta \text{FemaleCEO}_{it} + \sum_{q} \delta_q \text{PerfQuintile}_{q,it} + \sum_{q} \theta_q (\text{FemaleCEO}_{it} \times \text{PerfQuintile}_{q,it}) + \gamma' X_{it}\right). \quad (7)$$

Interpretation: The turnover tests mirror the compensation tests. The core question is whether poor performance leads to a stronger probability of departure for female CEOs.

3.8 Board representation and background tests

The board tests use regressions of female representation and compensation on gender composition of boards and leadership teams. The background tests compare education and career histories by gender.

Interpretation: These are mechanism tests. They ask whether observed pay and turnover differences can be explained by governance composition or observable CEO background differences.

4 Identification and empirical design

4.1 What is identified cleanly

- The paper cleanly documents conditional associations between CEO gender and pay, incentives, turnover, and hospital policies.
- The crisis design is the strongest quasi-experimental component because it uses a large unexpected shock and matched hospitals.
- Hospital fixed effects help separate a CEO-gender effect from permanent hospital differences.

Interpretation: The paper is strongest when multiple empirical designs point in the same direction: no policy differences, but clear compensation and turnover differences.

4.2 What remains difficult

- CEOs are not randomly assigned to hospitals.
- Boards may have private information about CEO ability that the researcher does not observe.
- Performance metrics may not fully capture CEO effort, mission quality, patient outcomes, or internal management.
- Gender differences in background may occur earlier in careers, before the CEO appointment, and therefore may not be fully observed in late-career data.

Interpretation: The paper therefore avoids claiming that boards are definitively biased. Instead, it documents board behavior that is consistent with a perception that female CEOs are less productive or with unobserved differences that boards use but researchers cannot observe.

4.3 Why the null decision-making results matter

- The null results are not merely insignificant correlations; the estimates are generally small and stable across designs.
- They weaken the claim that women in top executive roles manage firms differently because of risk aversion or benevolence.
- They shift attention from female managerial style to board evaluation and labor-market treatment.

Interpretation: The paper's main contribution is the contrast between "no difference in decisions" and "large difference in how CEOs are paid and replaced."

5 Tables and figures

5.1 Figure 1 and Table I: Female leadership and sample description

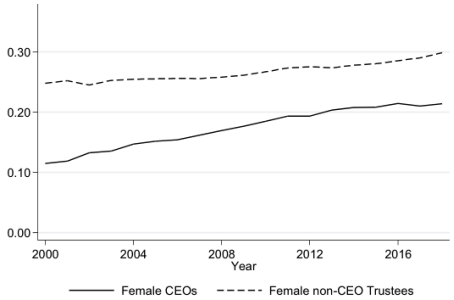
Read:

- Figure 1 shows that the share of female hospital CEOs rises from roughly 11% in 2000 to about 21% in 2018.
- Female non-CEO trustees are more common throughout, rising from about 25% to roughly 30%.
- Table I shows that female CEOs lead smaller hospitals: female-led hospitals have mean service revenue of about $158 million versus $216 million for male-led hospitals.
- Female- and male-led hospitals look similar on profit margin, leverage, cash/assets, securities/assets, Medicare ratio, Medicaid ratio, and charity care.

Interpretation: The descriptive statistics already foreshadow the paper's main split. Female CEOs are more common than in public-company samples, which makes the study feasible. But the gender difference is mostly about organizational scale, not obvious differences in risk or mission intensity.

Economic message: If women were systematically selected into more altruistic or safer hospitals, Table I should show large differences in charity care, Medicaid patients, investment, or leverage. It does not.

What not to over-read: Table I is descriptive. It does not prove that women and men would choose the same policies if randomly assigned to identical hospitals.



Fraction of female CEOs and trustees in U.S. hospitals. The figure
female CEOs and female non-CEO trustees by year for 25,762 hospital-y
from 2000 to 2018.

Figure 1: Female CEOs and non-CEO trustees.

Table I
Descriptive Statistics

The table presents descriptive statistics for 25,762 hospital-years (1,981 hospitals) from 2000 t
2018. Variable definitions are in the Appendix.

Panel A: Descriptive Statistics for the Full Sample

	Mean	Median	Std	P5	P95	*N*
Service Revenues ($ mil.)	205.642	103.940	301.608	10.343	711.659	25,762
Assets ($ mil.)	253.205	106.463	493.724	7.408	901.876	25,762
Profit Margin	0.038	0.036	0.072	−0.079	0.160	25,544
Investment	0.056	0.005	0.191	−0.113	0.416	25,061
Growth in Services	0.060	0.056	0.106	−0.081	0.210	25,532
Contributions	0.019	0.006	0.037	0.000	0.084	25,534
Leverage	0.232	0.220	0.191	0.000	0.579	24,676
Cash/Assets	0.032	0.003	0.055	0.000	0.155	24,756
Securities/Assets	0.220	0.197	0.168	0.005	0.537	13,597
System	0.457	0.000	0.498	0.000	1.000	25,762
Density Decile HSA	3.752	3.000	3.010	1.000	10.000	25,730
Competition Rank	1.570	1.000	0.836	1.000	3.000	25,685
Medicare Ratio	0.504	0.523	0.183	0.110	0.774	25,753
Medicaid Ratio	0.203	0.163	0.156	0.033	0.572	25,762
Charity Care	0.014	0.009	0.016	0.001	0.048	9,688

Panel B: Descriptive Statistics by CEO Gender

	Female CEOs			Male CEOs		
	Mean	Median	*N*	Mean	Median	*N*
Service Revenues ($ mil.)	158.359	66.832	4,456	215.531	112.160	21,306
Assets ($ mil.)	201.609	73.427	4,456	263.996	116.137	21,306
Profit Margin	0.037	0.033	4,414	0.038	0.036	21,130
Investment	0.056	−0.001	4,361	0.056	0.007	20,700
Growth in Services	0.053	0.048	4,410	0.061	0.058	21,122
Contributions	0.023	0.006	4,408	0.018	0.006	21,126
Leverage	0.221	0.197	4,193	0.234	0.224	20,483
Cash/Assets	0.033	0.003	4,209	0.032	0.004	20,547
Securities/Assets	0.225	0.190	2,201	0.219	0.198	11,396
System	0.477	0.000	4,456	0.452	0.000	21,306
Density Decile HSA	3.773	2.000	4,448	3.748	3.000	21,282
Competition Rank	1.630	1.000	4,440	1.558	1.000	21,245
Medicare Ratio	0.503	0.525	4,456	0.504	0.523	21,297
Medicaid Ratio	0.206	0.165	4,456	0.202	0.162	21,306
Charity Care	0.014	0.008	1,950	0.015	0.009	7,738

Table I: Descriptive statistics.

5.2 Tables II and III: Do female CEOs match with different hospital policies?

Read:

- Table II studies hospital financing and investment. The dependent variable is whether the CEO is female.
- The results do not show that female CEOs are associated with lower leverage, more cash, higher securities holdings, or lower investment.
- Table III studies patient mix and stakeholder-related outcomes.
- The coefficients do not support the view that female CEOs systematically tilt hospitals toward less profitable or more vulnerable patients.

Interpretation: These tables test the first-order prediction from gender-preference stories. If women are more risk averse, female CEOs should be linked to safer financing or investment. If women are more benevolent, they should be linked to a higher Medicaid ratio, more charity care, or stakeholder-friendly personnel policies. The tables do not provide that evidence.

Economic message: The null result is important because it separates top executive behavior from findings in lower-level executive or board studies. At the CEO level in hospitals, observed policies are not visibly gendered.

What not to over-read: These regressions are not causal assignment tests. They show that observed matching and policy patterns are inconsistent with large, systematic differences in management style.

Table II
Female CEOs and Hospital Financing and Investment

The table presents results from OLS regressions of an indicator for female CEO on hospital characteristics. Variable definitions are in the Appendix. The sample in each column consists of the hospital-year panel described in Table I, constrained to observations with nonmissing independent variables. Standard errors (in parentheses) are clustered at the hospital level. *, **, and *** indicate p-values of less than 0.1, 0.05, and 0.01.

	Dependent Variable = 1 if CEO is Female						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Leverage	0.020	0.025	0.053	0.062			
	(0.031)	(0.031)	(0.043)	(0.043)			
Cash / Assets	−0.150	−0.092	−0.035	0.005			
	(0.096)	(0.076)	(0.151)	(0.101)			
Securities / Assets			0.071	0.039			
			(0.048)	(0.045)			
Investment					0.010	0.013	
					(0.015)	(0.011)	
Competition Rank							0.033***
							(0.010)
Log(Service Rev.)	−0.053***	−0.039**	−0.061***	−0.053**	−0.055***	−0.048***	−0.057***
	(0.006)	(0.018)	(0.009)	(0.023)	(0.007)	(0.017)	(0.007)
Profit Margin	0.123*	0.061	0.067	0.043	0.121*	0.074	0.113
	(0.072)	(0.053)	(0.093)	(0.061)	(0.073)	(0.049)	(0.072)
Density Decile HSA	0.012***		0.012***		0.014***		0.009***
	(0.003)		(0.004)		(0.003)		(0.003)
Year FE	Y	Y	Y	Y	Y	Y	Y
Hospital FE	N	Y	N	Y	N	Y	N
N	23,611	23,627	12,205	12,152	25,020	25,031	25,468

Table II: Female CEOs and hospital financing/investment.

Female CEOs and Hospital Tilt Toward Less Profitable Patients

The table presents results from OLS regressions of an indicator for female CEO on hospital characteristics. Variable definitions are in the Appendix. The sample in each column consists of the hospital-year panel described in Table I, constrained to observations with nonmissing independent variables. Standard errors (in parentheses) are clustered at the hospital level. *, **, and *** indicate p-values of less than 0.1, 0.05, and 0.01.

	Dependent Variable = 1 if CEO is Female							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Pct. Unemployed	−0.000							
	(0.002)							
Log(Median Inc.)		−0.003						
		(0.032)						
Medicare Ratio			−0.038	0.039				
			(0.049)	(0.034)				
Medicaid Ratio			−0.031	0.037				
			(0.056)	(0.042)				
Charity Care					−0.160	−0.175		
					(0.500)	(0.408)		
Contributions							0.159	0.224*
							(0.167)	(0.126)
Profit Margin	0.105	0.107	0.102	0.047	0.023	0.115*	0.098	0.027
	(0.071)	(0.071)	(0.071)	(0.049)	(0.105)	(0.064)	(0.072)	(0.051)
Log(Service Rev.)	−0.053***	−0.053***	−0.054***	−0.033*	−0.048***	−0.044*	−0.052***	−0.029
	(0.007)	(0.007)	(0.007)	(0.018)	(0.009)	(0.026)	(0.007)	(0.018)
Density HSA	0.013***	0.013***	0.013***		0.008**		0.013***	
	(0.003)	(0.003)	(0.003)		(0.004)		(0.003)	
Year FE	Y	Y	Y	Y	Y	Y	Y	Y
Hospital FE	N	N	N	Y	N	Y	N	Y
N	25,513	25,513	25,504	25,523	9,642	9,567	25,291	25,306

Table III: Female CEOs and less profitable patients.

5.3 Figure 2 and Table IV: The financial crisis setting and matched samples

Read:

- Figure 2 shows a broad hospital-sector shock around the 2008 financial crisis.
- Securities/assets fall sharply in 2008, while profit margin and investment fall in 2009.
- Personnel growth, salary growth, service growth, and investment all weaken around the crisis.
- Table IV shows that the female-CEO treatment sample and male-CEO matched controls are closely balanced in 2007.

Interpretation: Figure 2 motivates the shock design: hospitals were affected materially by the crisis. Table IV then shows why the comparison is credible: female- and male-led hospitals are similar immediately before the shock.

Economic message: If gender differences in CEO risk aversion or altruism matter most under stress, the crisis should reveal them. The paper uses this setting as a high-stakes test.

What not to over-read: The matching sample is not a randomized experiment, but it reduces concern that female CEOs simply run very different hospitals before the shock.

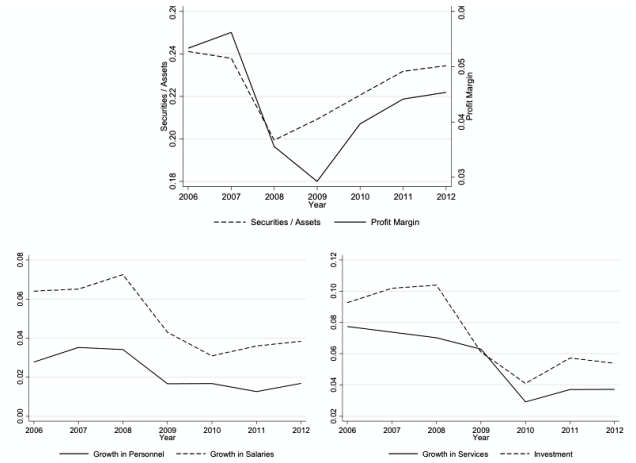


Figure 2. Hospital performance, investment, salary growth, and personnel growth around the 2008 financial crisis. The panels show averages by year. Variable definitions are

Figure 2: Hospital outcomes around the crisis.

Table IV
Matched Samples of Hospitals with Male and Female CEOs

The table presents descriptive statistics for 134 hospitals with female CEOs in 2007 and a matched sample of 271 hospitals with male CEOs. The table reports weighted means and medians for the matched and treated samples in 2007. Variable definitions are in the Appendix.

	Treatment Sample (N = 134)		Control Sample (N = 271)	
	Mean	Median	Mean	Median
Service Rev. ($ mil.)	128.508	77.680	125.301	72.419
Assets ($ mil.)	150.367	74.571	164.524	76.820
Profit Margin	0.057	0.045	0.053	0.038
Investment	0.152	0.052	0.151	0.045
Growth in Services	0.068	0.063	0.061	0.055
Contributions	0.021	0.008	0.027	0.009
Leverage	0.254	0.251	0.266	0.259
Cash/Assets	0.033	0.001	0.031	0.003
Securities/Assets	0.233	0.176	0.232	0.207
System	0.366	0.000	0.363	0.000
Density Decile HSA	4.045	2.000	3.794	2.000
Competition Rank	1.679	1.000	1.473	1.000
Medicare Ratio	0.515	0.543	0.472	0.477
Medicaid Ratio	0.206	0.140	0.234	0.204

Table IV: Matched samples.

5.4 Table V and Figure 3: Do male and female CEOs respond differently to the crisis?

Read:

- Table V reports the crisis difference-in-differences results.
- The interaction $Post_Crisis \times F$ is small and statistically insignificant across revenue growth, profit margin, investment, personnel growth, salary growth, and Medicaid ratio.
- The female-CEO sample and male-CEO control sample both experience large post-crisis declines.
- Figure 3 shows year-by-year gender interactions around the crisis. The coefficients are centered near zero and confidence intervals generally include zero.

Interpretation: This is one of the strongest pieces of evidence in the paper. The crisis was large, unexpected, and financially relevant. Yet female-led hospitals do not appear to protect employees more, preserve investment differently, take more Medicaid patients, or respond more conservatively.

Economic message: The result weakens the argument that female CEO preferences drive materially different corporate policies. Under a major shock, the organizational response is similar across CEO gender.

What not to over-read: A null crisis response does not prove that gender never matters. It means the paper does not detect meaningful differences in these hospital outcomes during this particular shock.

Table V

Hospital Response to the Financial Crisis as a Function of CEO Gender

The table presents results from OLS regressions of outcome variables listed in column headings. The sample consists of treatment and control hospitals described in Table IV from 2006 through 2011. The *Post_Crisis* dummy is set to one for years 2009 through 2011. The treatment sample ($F = 1$) includes 134 hospitals with female CEOs in 2007 through at least 2010. The matched control sample includes 271 hospitals with male CEOs in 2007 through at least 2010. Variable definitions are in the Appendix. Standard errors (in parentheses) are clustered at the hospital level. *, **, and *** indicate p-values of less than 0.1, 0.05, and 0.01.

Dependent Var.:	Growth in Services	Profit Margin	Investment	Growth in Personnel	Growth in Salaries	Medicaid Ratio
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Full Sample						
Post_Crisis # F	−0.001	−0.003	−0.002	−0.003	−0.012	−0.000
	(0.008)	(0.006)	(0.028)	(0.007)	(0.010)	(0.008)
Hospital FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
N	2,285	2,286	2,277	2,251	2,285	2,287
Panel B: Treatment Sample (Female CEOs; F = 1)						
Post_Crisis	−0.030***	−0.016***	−0.072***	−0.029***	−0.029***	−0.001
	(0.007)	(0.005)	(0.020)	(0.005)	(0.008)	(0.006)
Hospital FE	Y	Y	Y	Y	Y	Y
N	754	754	753	748	753	755
Panel C: Control Sample (Male CEOs; F = 0)						
Post_Crisis	−0.029***	−0.013***	−0.069***	−0.026***	−0.017**	−0.001
	(0.004)	(0.004)	(0.019)	(0.005)	(0.007)	(0.005)
Hospital FE	Y	Y	Y	Y	Y	Y
N	1,531	1,532	1,524	1,503	1,532	1,532

Columns (1) to (3) and Figure 3 show that the treatment and control groups'

Table V: Crisis response by CEO gender.

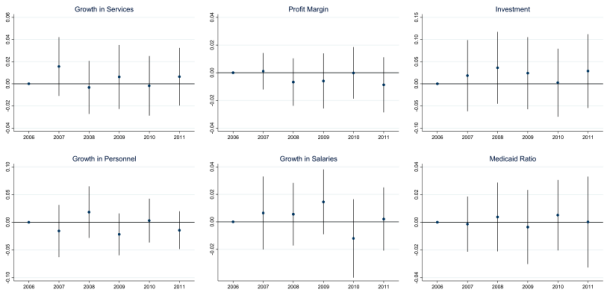


Figure 3. The effects of CEO gender on hospital outcomes around the financial crisis. The panels show regression coefficients and standard errors from regressions of hospital outcomes on CEO gender interacted with year dummies. The regressions are similar to those in Table V, Panel A, except that the *Female CEO* is interacted with year indicators for the years around the financial crisis (rather than with the *Post_Crisis* dummy). Variable definitions are in the Appendix. (Color figure can be viewed at wileyonlinelibrary.com)

discussion in Matsa and Miller (2013)). In columns (4) to (6), I test whether

Figure 3: Event-study crisis effects.

5.5 Tables VI and VII: Descriptive pay gap and regression-adjusted pay gap

Read:

- Table VI shows a large raw compensation difference: female CEOs average about $457,000 versus $609,000 for male CEOs.
- Female CEOs have similar age, slightly shorter tenure, and are less likely to hold multiple related positions.
- Table VII shows that the raw pay gap is about 32%.
- Controlling for hospital size, profitability, growth, system status, tenure, age, and hospital fixed effects reduces the gap, but it remains about 7.8% in the strictest full-sample specification.

Interpretation: The key lesson is that hospital size explains much of the pay gap, but not all of it. The fixed-effect specification is especially informative because it compares pay within the same hospital across CEOs and still finds a residual gap.

Economic message: The compensation results imply that gender differences appear in labor-market outcomes even when they do not appear in hospital policy choices.

What not to over-read: The residual pay gap is not automatically proof of discrimination. It could reflect bargaining, unobserved responsibilities, unobserved CEO quality, or board beliefs. But it is difficult to explain using the policy-choice evidence alone.

Table VI

Descriptive Statistics for CEO Pay and Turnover

The table presents descriptive statistics for CEO attributes by CEO gender. *CEO Pay* (in $thousands) is reported separately for the pre-2008 and post-2008 periods due to a change in the IRS reporting requirements (see details and other variable definitions in the Appendix).

	Female CEOs			Male CEOs		
	Mean	Median	*N*	Mean	Median	*N*
CEO Pay (all years)	456.935	299.643	3,561	609.080	424.814	16,959
CEO Pay (pre-2008)	330.534	250.055	1,083	447.304	349.777	6,891
CEO Pay (post-2008)	512.178	332.463	2,478	719.807	503.553	10,068
Tenure	4.569	3.000	4,456	4.915	4.000	21,306
CEO Age	54.967	56.000	3,937	55.133	56.000	18,879
Multiple Positions	0.087	0.000	4,456	0.112	0.000	21,306
Turnover	0.110	0.000	4,237	0.106	0.000	20,500
Turnover ("forced")	0.071	0.000	3,733	0.064	0.000	18,149

to total reportable compensation, which includes base salary and bonus (see

Table VI: Pay and turnover statistics.

Table VII

Gender Gap in CEO Pay

The table presents results from OLS regressions of the logarithm of CEO pay on an indicator for f male CEOs and controls. The time-varying hospital controls are lagged. The top panel includes th full sample spanning years 2000 through 2018, constrained to observations with nonmissing dat Panel B includes only the period with consistent IRS reporting of CEO compensation (see Sectio VI.A). Variable definitions are in the Appendix. Standard errors (in parentheses) are clustered the hospital level. *, **, and *** indicate *p*-values of less than 0.1, 0.05, and 0.01.

Panel A: Full-Sample Regressions

	Dependent Variable = Log (CEO Pay)				
	(1)	(2)	(3)	(4)	(5)
Female CEO	−0.322***	−0.125***	−0.126***	−0.133***	−0.078**
	(0.037)	(0.022)	(0.020)	(0.020)	(0.032)
Log(Service Revenues)		0.444***	0.372***	0.364***	0.081**
		(0.009)	(0.010)	(0.010)	(0.034)
Profit Margin			0.568***	0.556***	0.503***
			(0.106)	(0.107)	(0.088)
Growth in Services			−0.123**	−0.083	−0.025
			(0.054)	(0.051)	(0.044)
Contributions			0.852***	0.812***	−0.280
			(0.222)	(0.221)	(0.207)
Density Decile HSA			0.047***	0.048***	
			(0.004)	(0.004)	
System			0.063***	0.063***	
			(0.016)	(0.016)	
CEO Tenure			0.029***	0.027***	0.029***
			(0.002)	(0.002)	(0.003)
Multiple Positions			0.488***	0.507***	0.334***
			(0.036)	(0.034)	(0.039)
Year FE	Y	Y	Y	Y	Y
Hospital FE	N	N	N	N	Y
CEO Age Dummies	N	N	N	Y	Y
N	20,520	18,498	18,143	16,271	16,193

Panel B: Regressions by Period with Consistent Reporting (Post-2008)

	Dependent Variable = Log (CEO Pay)			
	2009 to 2013		2014 to 2018	
	(1)	(2)	(3)	(4)
Female CEO	−0.345***	−0.121***	−0.352***	−0.107**
	(0.045)	(0.028)	(0.052)	(0.028)
Log(Service Revenues)		0.423***		0.442***
		(0.012)		(0.011)
Controls	N	Y	N	Y
Year FE	Y	Y	Y	Y
CEO Age Dummies	N	Y	N	Y
N	6,743	5,574	5,803	4,900

Table VII: Gender gap in CEO pay.

5.6 Table VIII and Figure 4: Pay-for-performance sensitivity

Read:

- Table VIII estimates compensation as a function of performance quintile.
- Male CEO pay increases strongly with performance rank: the coefficient for the best quintile is about 0.150 in the male-CEO column.
- Female CEO pay is much flatter: the best-quintile coefficient is about 0.053 and is not statistically strong.
- The female interaction with the best quintile is negative and statistically significant in combined regressions.
- Figure 4 visualizes the result: male CEO pay rises monotonically with performance, while female CEO pay is flatter and noisier.

Interpretation: This is not simply a level pay gap. It is an incentives gap. Boards appear to give male CEOs stronger financial rewards for good performance, while female CEOs receive weaker upside incentives.

Economic message: The pay-for-performance result is hard to reconcile with the earlier null policy results if the only story is that women manage differently. The board's compensation function itself appears gendered.

What not to over-read: Flatter pay-for-performance need not mean female CEOs work less hard. It can reflect board beliefs, bargaining, contract design, risk-sharing preferences, or unobserved differences in CEO roles.

Table VIII

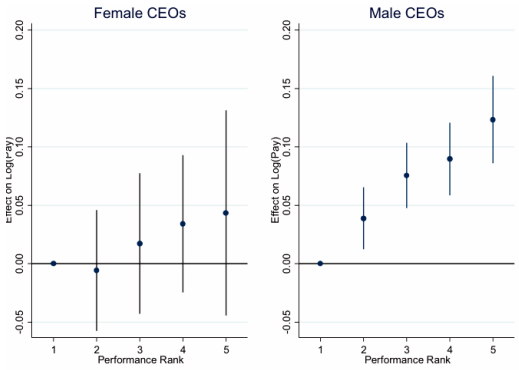
Pay-for-Performance Sensitivity

The table presents results from OLS regressions with the logarithm of CEO pay as the depender variable. *Female CEO* is a dummy equal to one when the hospital CEO is female. *Perf_Quintile* to *Perf_Quintile5* are quintile indicators for the hospital's financial performance measured as i profit margin lagged by one year relative to pay (*Perf_Quintile1* is excluded; details are in the Ap pendix). Controls include *Log(Service Revenue)*, *Growth in Services*, *Contributions*, *System*, *Tenur Multiple Positions*. The variables are lagged by one year with respect to pay. Variable definitior are in the Appendix. Standard errors (in parentheses) are clustered at the hospital level. *, **, an *** indicate p-values of less than 0.1, 0.05, and 0.01.

	Dependent Variable = Log (CEO Pay)				
	All CEOs (1)	Female (2)	Male (3)	All CEOs (4)	All CEO (5)
Female CEO	−0.065**			−0.017	−0.029
	(0.030)			(0.045)	(0.049)
Perf_Quintile2	0.043***	0.020	0.050***	0.050***	0.037***
	(0.013)	(0.024)	(0.014)	(0.014)	(0.014)
Perf_Quintile3	0.076***	0.021	0.088***	0.087***	0.074***
	(0.013)	(0.029)	(0.015)	(0.015)	(0.015)
Perf_Quintile4	0.088***	0.052*	0.101***	0.096***	0.084***
	(0.015)	(0.028)	(0.017)	(0.017)	(0.016)
Perf_Quintile5 (Best)	0.134***	0.053	0.150***	0.150***	0.126***
	(0.019)	(0.041)	(0.021)	(0.021)	(0.020)
Female CEO # Perf_Quintile2				−0.040	−0.046
				(0.037)	(0.039)
Female CEO # Perf_Quintile3				−0.064	−0.060
				(0.039)	(0.041)
Female CEO # Perf_Quintile4				−0.043	−0.044
				(0.044)	(0.044)
Female CEO # Perf_Quintile5				−0.096**	−0.105**
				(0.046)	(0.048)
Year FE	Y	Y	Y	Y	Y
Hospital FE	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y
CEO Age Dummies	N	N	N	N	Y
N	18,066	3,047	14,934	18,066	16,193

5% level. As illustrated in Figure 4, the gap in pay between male and femal

Table VIII: Pay-for-performance sensitivity.



4. Pay-for-performance sensitivity for female and male CEOs. The panels on coefficients and standard errors from regressions of the logarithm of CEO pay or

Figure 4: Pay-performance slopes.

5.7 Table IX and Figure 5: Turnover-performance sensitivity

Read:

- Table IX estimates CEO turnover as a function of performance quintile.
- Poor performance predicts turnover for both male and female CEOs.
- The poor-performance effect is larger for female CEOs: the worst-quintile coefficients for women are much larger than for men.
- Figure 5 shows that the turnover probability for female CEOs falls steeply as performance rank improves.
- The gender gap in turnover is concentrated at low performance; high-performing men and women look more similar.

Interpretation: Table IX and Figure 5 mirror Table VIII and Figure 4 in an asymmetric way. Female CEOs receive weaker rewards for good performance but stronger punishment for poor performance.

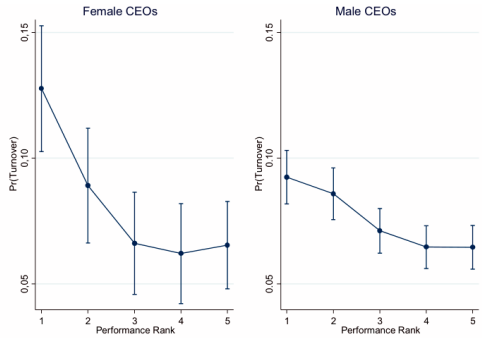
Economic message: This combination suggests that boards may interpret negative performance signals more harshly for women than for men, while giving women less performance-based upside.

What not to over-read: The paper cannot fully distinguish biased updating from unobserved productivity differences. But the asymmetry is economically meaningful because it appears in both pay contracts and CEO exits.

(left panel) or any turnover (right panel). Forced turnover is defined as turnover of CEOs aged 60 or younger. *Perf_Quintile1* (*Worst*) to *Perf_Quintile5* (*Best*) are quintile indicators for the hospital's financial performance measured as its profit margin in the year prior to turnover (*Perf_Quintile5* is excluded; details are in the Appendix). Columns (4) and (8) include interaction terms of *Female CEO* with *Perf_Quintile1* (*Worst*). *Marginal Effects* are computed as the difference in marginal effects of *Perf_Quitile1* for *Female CEO* = 1 versus *Female CEO* = 0 with other variables set to their actual values. Controls include *Log(Service Revenue)*, *Growth in Services*, *Contributions*, *System*, *Tenure*, and *Multiple Positions*. The variables are measured in the year prior to turnover. Variable definitions are in the Appendix. Standard errors (in parentheses) are clustered at the hospital level. *, **, and *** indicate *p*-values of less than 0.1, 0.05, and 0.01.

	Dependent Variable = 1 for Turnover Years							
	"Forced" Turnover (Sample with Age Data)				All Turnover (Full Sample)			
	All CEOs (1)	Female (2)	Male (3)	All CEOs (4)	All CEOs (5)	Female (6)	Male (7)	All CEOs (8)
Coefficients								
Female CEO	0.030			−0.017	0.003			−0.030
	(0.037)			(0.043)	(0.030)			(0.034)
Perf_Quintile1 (Worst)	0.239***	0.391***	0.199***	0.201***	0.266***	0.437***	0.227***	0.240***
	(0.044)	(0.098)	(0.050)	(0.048)	(0.035)	(0.082)	(0.039)	(0.038)
Perf_Quintile2	0.159***	0.172*	0.156***	0.159***	0.165***	0.266***	0.145***	0.165***
	(0.044)	(0.104)	(0.050)	(0.044)	(0.035)	(0.085)	(0.040)	(0.035)
Perf_Quintile3	0.043	0.006	0.052	0.042	0.089**	0.100	0.084**	0.088**
	(0.045)	(0.112)	(0.050)	(0.045)	(0.036)	(0.092)	(0.039)	(0.036)
Perf_Quintile4	0.000	−0.029	0.001	−0.001	0.034	0.111	0.016	0.034
	(0.046)	(0.111)	(0.051)	(0.046)	(0.037)	(0.090)	(0.040)	(0.037)
Female CEO # Perf_Quintile1 (Worst)				0.192**				0.137**
				(0.083)				(0.066)
Marginal Effects								
Female CEO # Perf_Quintile1 (Worst)				0.036**				0.032**
				(0.015)				(0.016)
Year FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y
CEO Age Dummies	Y	Y	Y	Y	N	N	N	N

Table IX: CEO turnover regressions.



ıre 5. Turnover-performance sensitivity for female and male CEOs. The panels
ginal effects and standard errors estimated from probit regressions of CEO turnover. T
sions estimate the probability of "forced" CEO turnover, defined as departures of CEOs a

Figure 5: Turnover-performance slopes.

5.8 Table X: Female board representation and CEO gender

Read:

- Table X studies whether female boards are associated with female CEOs and female non-CEO executives.
- Female directors are positively related to female CEO incidence in some specifications.
- Female directors are also related to the presence of female non-CEO executives.
- Panel B asks whether female boards reduce the CEO pay gap. The gender pay gap remains negative even when board gender is included.

Interpretation: Board gender composition may influence the pipeline or representation of women, but it does not obviously eliminate the pay gap at the CEO level in this sample.

Economic message: The result limits a simple governance-diversity explanation. Adding women to boards may matter for representation, but the compensation asymmetry persists.

What not to over-read: Table X is a mechanism test with smaller and more selected samples. It should not be read as a definitive estimate of the causal effect of female trustees.

Female Representation on Hospital Boards and CEO Gender

Panel A presents results from regressions of an indicator for *Female CEO* (or the fraction of *Female Executives*). The sample consists of the hospital-year panel described in Table I and Internet Appendix Table IA.IX, constrained to observations with nonmissing data. *Female Executives: Non-CEO, Non-Director* is the fraction of women among the non-CEO and nondirector executives. The independent variables are lagged. *Female Directors: Non-CEO* is the fraction of women among non-CEO directors. *Board Size* and *Independent Directors* are the number of directors and independent directors (the latter available after 2008). Controls include lagged *Log(Service Revenue)*, *Growth in Services*, *Contributions*, *Profit Margin*, and (in regressions without hospital fixed effects) *System* and *Density HSA*. Panel B presents results from regressions of *Log(CEO Pay)*. The independent variables are lagged. *Female Board Dummy* is an indicator for the fraction of female non-CEO directors above the sample median of 25%. The regressions in columns (1) and (2) are run on subsamples split on lagged *Female Board Dummy*. In addition to the controls in Panel A, the controls include *Tenure* and *Multiple Positions*. Variable definitions are in the Appendix. Standard errors (in parentheses) are clustered at the hospital level. *, **, and *** indicate p-values of less than 0.1, 0.05, and 0.01.

Panel A: Frequency of Female CEOs or Female Top Executives

Dep. Variables:	Female CEO				Female Executives: Non-CEO, Non-Director			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Female Directors: Non-CEO	0.136**	0.146***	−0.040	0.011	0.180***	0.123***	0.016	−0.021
	(0.060)	(0.043)	(0.053)	(0.034)	(0.055)	(0.033)	(0.054)	(0.032)
Log(Board Size)	−0.016	0.002	−0.020	0.009	−0.022	−0.014	−0.031	0.008
	(0.039)	(0.015)	(0.046)	(0.011)	(0.042)	(0.011)	(0.040)	(0.012)
Log(Independent Directors)	−0.004		−0.010		−0.007		−0.010	
	(0.030)		(0.028)		(0.033)		(0.024)	
Hospital FE	N	N	Y	Y	N	N	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y
N	10,199	20,549	10,089	20,480	9,133	18,613	8,997	18,522

(*Continued*)

Table X, Panel A: Female representation and CEO gender.

Table X—*Continued*

Panel B: Gender Gap in CEO Pay

	Dependent Variable = Log(CEO Pay)			
	Female Board Dummy = 1	Female Board Dummy = 0	Full Sample	
	(1)	(2)	(3)	(4)
Female CEO	−0.107***	−0.170***	−0.161***	−0.080**
	(0.022)	(0.031)	(0.031)	(0.038)
Female CEO #Female Board Dummy			0.049	0.007
			(0.033)	(0.029)
Female Board Dummy			−0.037**	−0.002
			(0.015)	(0.014)
Hospital FE	N	N	N	Y
Year FE	Y	Y	Y	Y
Controls	Y	Y	Y	Y
CEO Age Dummies	Y	Y	Y	Y
N	8,543	7,422	15,965	15,884

Table X, Panel B: Board gender and CEO pay gap.

5.9 Table XI: CEO educational and career backgrounds

Read:

- Female CEOs are much more likely to have undergraduate nursing backgrounds: 41.0% versus 3.3% for male CEOs.
- Male CEOs are more likely to have undergraduate business backgrounds and graduate business degrees.
- Female CEOs are more often internally promoted: 43.8% versus 33.6%.
- Among external hires, female and male CEOs have broadly similar senior-executive histories, though women are somewhat less likely to come from hospital senior-executive roles.

Interpretation: Table XI shows that gender differences in backgrounds are real, especially early in careers. But by the time CEOs are appointed, both men and women generally have senior executive experience.

Economic message: Background differences may explain part of board evaluation or compensation, but they do not directly explain why female CEOs show similar hospital policy choices yet receive lower pay and face steeper turnover penalties.

What not to over-read: Background data are from smaller manually collected subsamples. They are useful for interpretation, not a complete decomposition of the gender pay and turnover gaps.

Table XI

CEO Educational and Career Backgrounds

The table presents descriptive statistics for educational and career backgrounds by gender for subsamples of CEOs (samples construction and data definitions are in Internet Appendix Section IV). Panel A shows statistics on educational background for 984 CEOs (229 of which are female) with available data. Panel B shows the fraction of CEOs promoted internally for 2,202 CEOs (486 of which are female) for which internal promotion versus outside hires could be determined. For the internally promoted CEOs, the table also shows statistics for the most common previous titles held by the CEOs. Panel C shows statistics on previous occupations for a matched sample of 441 externally hired CEOs (133 of which are women). The occupations are for the most recent job held before the CEO appointment. Panel C also reports the fraction of previous senior executives whose previous employer was a hospital, the number of years in the previous position, and the fraction of externally hired CEOs who were poached from a CEO position held at another hospital in the main sample (details are in Internet Appendix Section IV).

	Female CEOs	Male CEOs
Panel A: Educational Background		
Undergraduate Major: Business	0.332	0.490
Undergraduate Major: Nursing	0.410	0.033
Graduate Degree: Business	0.747	0.866
Panel B: Internal Promotions		
Fraction of Internal Promotions	0.438	0.336
Previous Title: Vice President	0.340	0.314
Previous Title: COO	0.225	0.207
Previous Title: CFO	0.094	0.114
Panel C: External Hires		
Occupation:		
Senior Executive	0.872	0.909
Medical	0.015	0.010
Other	0.053	0.032
Fraction of Senior Executives at Hospitals	0.828	0.896
Average Tenure in Previous Position (Number of Years)	5.469	5.358
Fraction of CEOs Poached from a Sample Hospital	0.046	0.060

Table XI: CEO educational and career backgrounds.

6 Short summary

- The paper studies female CEOs in U.S. nonprofit hospitals, where women are much more common than in public-company CEO samples.
- The share of female CEOs rises over the sample period, but women remain a minority of CEOs.
- Female CEOs tend to lead smaller hospitals, but not hospitals with obviously safer financing, lower investment, or more charitable patient policies.
- OLS and fixed-effect evidence does not show a distinct female management style.
- Matched crisis tests show that female- and male-led hospitals respond similarly to the 2008 financial crisis.
- Female CEOs earn less than male CEOs; the raw gap is large and the controlled gap remains statistically meaningful.
- Female CEOs have flatter pay-for-performance incentives.
- Female CEOs face stronger turnover consequences after poor financial performance.
- Female board representation is related to female leadership in some specifications, but it does not eliminate the CEO pay gap.
- Female CEOs have different early educational backgrounds, especially nursing, but broadly similar senior executive backgrounds before CEO appointment.
- The paper's central contrast is: no detectable gender difference in hospital policies, but clear gender differences in pay, incentives, and turnover.

7 Conclusion

7.1 Core conclusion

Lewellen finds no evidence that female hospital CEOs manage hospitals differently from male CEOs in ways predicted by common gender-preference theories. Female CEOs do not choose safer financing, reduce investment, expand charity care, protect employees more during crisis, or shift hospitals toward less profitable patients. Yet female CEOs are paid less, receive weaker pay-for-performance incentives, and face stronger turnover penalties after poor performance.

7.2 Why the null management-style result matters

The absence of detectable policy differences is a central contribution. Much of the gender and corporate-governance literature suggests that women may be more risk averse, less overconfident, more stakeholder-oriented, or more benevolent. If those traits were important at the CEO level in this setting, they should appear in leverage, cash, investment, salaries, staffing, Medicaid share, or charity care. The paper does not find such patterns.

This result is particularly important because hospitals are mission-oriented nonprofit organizations. If benevolence or stakeholder orientation were going to show up in CEO decisions, hospitals are a natural place to look. The fact that the paper does not find large differences suggests that top executives may face organizational constraints, board oversight, regulation, competitive pressures, or professional norms that dampen individual preference differences.

7.3 Why the compensation evidence matters

The pay results show that the gender gap is not only about who becomes CEO. Even among those who reach the CEO position, female CEOs receive lower pay. Hospital size explains a large part of the raw gap, because

women tend to lead smaller hospitals, but it does not explain all of it. The residual gap remains after rich controls and hospital fixed effects.

The pay-for-performance evidence deepens the result. Female CEOs are not only paid less on average; their compensation responds less strongly to good financial performance. This means the incentive contract itself differs by gender. The paper therefore points to a governance mechanism: boards may structure rewards differently for male and female CEOs.

7.4 Why the turnover evidence is striking

The turnover evidence is striking because it points in the opposite direction from compensation. If female CEOs simply had lower-powered contracts because boards wanted to insure them more, one might expect turnover-performance sensitivity to be flatter as well. Instead, poor performance triggers stronger turnover penalties for women.

This creates an asymmetric evaluation pattern. Good performance produces weaker pay upside for female CEOs, but bad performance produces stronger exit pressure. That combination is consistent with harsher board updating about female CEO ability, but it could also reflect unobserved productivity differences, different assignment to difficult jobs, or different bargaining power.

7.5 How to reconcile no policy differences with pay and turnover differences

The paper's main tension can be reconciled in several ways.

- **Biased beliefs or stereotypes.** Boards may interpret the same performance signal differently by gender, especially when performance is poor.
- **Unobserved productivity.** Boards may observe information about CEO ability or constraints that the researcher does not observe.
- **Bargaining differences.** Female CEOs may bargain differently over pay or be less likely to threaten outside offers.
- **Different contract roles.** Female CEOs may be hired into roles with different authority, autonomy, or implicit expectations, even within similar hospitals.
- **Career pipeline differences.** Gender differences in early educational and career backgrounds may affect how boards perceive transferable business skills, even if senior-career experience looks similar.

The paper does not definitively choose among these mechanisms, but it shows that a simple story based on women making safer or more altruistic decisions is not enough.

7.6 Contribution to the literature

The paper contributes to corporate finance by studying CEO gender in a setting large enough to support detailed empirical analysis. It also contributes to labor economics by showing that gender differences in top-executive pay and turnover persist even in a sector where women are relatively common. Finally, it contributes to governance by showing that boards' compensation and dismissal decisions can be gender-asymmetric even when managerial policies are not.

7.7 Limitations

Several limitations matter for interpretation. The paper studies hospitals, not public firms, banks, startups, or government agencies. The nonprofit setting is useful because of data richness and female representation, but governance objectives differ from shareholder-owned firms. CEO gender is not randomly assigned, and boards

may hold private information about CEO quality. The paper also cannot fully measure patient outcomes, management culture, internal effort, or the full range of CEO responsibilities.

7.8 Final takeaway

The paper's strongest message is that the gender gap at the top may be less about women choosing different policies and more about how women are evaluated once they hold the top job. In the hospital setting, female CEOs look similar to male CEOs in decision-making but different in compensation and turnover outcomes. That distinction is crucial: increasing female representation among CEOs may not radically change organizational policies, but it may reveal and challenge persistent asymmetries in executive labor markets.